

Ferran Alía Castillejos

Data science and physical engineering, UPC Barcelona

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NOW

I simulate microscope optics with the Physics of Life group at TU Dresden for the summer, then go back to Barcelona for my fourth year at UPC. I want to work where physical modeling and machine learning meet.

EXPERIENCE

SINCE 2026

Contributor, **Towards Data Science**

Long-form articles since March 2026, on simulation, high-performance computing and machine learning. Most of them come out of the projects below.

2026

Research intern, **Physics of Life, TU Dresden**

Simulating electromagnetic propagation through specimen and objective, as virtual instrumentation for learned brightfield imaging.

JUN–AUG 2025

Machine learning intern, **MLCode**

Built an automated framework for evaluating and benchmarking retrieval-augmented generation systems.

SINCE 2022

Volunteer mentor

Computer science and robotics workshops: Linux, 3D printing, programming.

2022–2023

Robotics teacher, **Punt Multimèdia**

Taught Arduino, micro:bit, and 3D printing and modeling.

EDUCATION

SINCE 2023

BSc Data Science & Engineering, **UPC**

Entering the fourth year of five. 8.75/10 average across both degrees.

SINCE 2023

BSc Physical Engineering, **UPC**

Taken in parallel with the data science degree rather than after it, on the CFIS dual-degree program.

SELECTED PROJECTS

2026

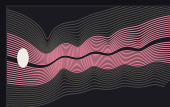


Aerodynamic optimization of a Formula 1 front wing

A surrogate model searched front-wing geometries on a fixed budget of solver runs rather than calling CFD at every step. The wing came out about 15% better on lift-to-drag.

Python • OpenFOAM • SLURM • MareNostrum V

2026



A physics-informed network for vortex shedding

A PINN with the Navier-Stokes residual inside its loss, trained against a high-fidelity OpenFOAM run, reproducing Kármán vortex shedding. A model fit to the data alone settles into a steady wake instead.

Python • PyTorch • OpenFOAM

2026



Persona adaptation in language models

A system prompt against three formats of fine-tuning data on Qwen3-4B-Instruct, all with LoRA at $r=16$, so the differences belong to the data.

Python • PyTorch • Transformers • LoRA



Predicting fetal distress from cardiotocography

Built with Hospital Sant Joan de Déu in a team of nine. Separate encoders over CTG traces, tabular clinical context and ultrasound, at around 0.78 AUC.
Python · PyTorch



Parallel N-body simulation

Direct $O(n^2)$ summation and a Barnes-Hut $O(n \log n)$ tree code in C++ with OpenMP. At $N = 10,000$ direct summation ran about $33\times$ faster on 112 threads than on one; the tree code never got past roughly $7\times$.
C++ · OpenMP



Parallel lattice Boltzmann solver

A D2Q9 solver with BGK collision and a Zou-He inlet in around 200 lines of C++: flow past a cylinder on a 400×400 lattice. Memory-bound, so how you move the data dominates the scaling.
C++ · OpenMP · MareNostrum 5



Fixed-wing bird flight simulation

A 3D Navier-Stokes solver written from scratch in NumPy, run on a bird's wing modeled in Blender and checked against OpenFOAM's icoFoam.
Python · NumPy · OpenFOAM · Blender



Fermi-Dirac statistics by Monte Carlo

A Markov chain over the microstates of a fermion system, recovering the Fermi-Dirac occupancy numerically rather than analytically.
Python · NumPy
Figures are schematic illustrations of each problem, not results.

SKILLS	LANGUAGES	PythonC++RHaskellMATLAB
	SCIENTIFIC	OpenFOAMOpenMPNumPyPyTorchBlenderCAD
	METHODS	CFDHPC & SLURMPhysics-informed MLMonte Carlo Surrogate modeling
	SPOKEN	Catalan (native)Spanish (native)English (professional)Arabic (beginner)

AWARDS	2024, 2025	Participant, Datathon FME
	2025	Physics summer school, University of Ljubljana
	SINCE 2023	Cum laude in several subjects
	2023	Admitted to CFIS on a grant, to read the two degrees at once. It takes a small cohort each year by its own entrance exam in mathematics and physics
	2019–2023	Graduated from high school with honors, and represented it at the Barcelona <i>Mostra de Recerca Jove</i> , after UPC's summer courses in algorithms and programming